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Sparse Estimation of Correlations among Microbiomes (SECOM)

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ABSTRACT

It is well-known that human gut microbiota form a complex ecosystem where microbes correlate with each other. Due to complex underlying interactions, some microbes may correlate nonlinearly. Yet, there are no measures in the literature that quantify these nonlinear relationships. We introduce a novel concept of correlation that allows for linear and nonlinear relationships among microbes. It is well-known that infant gut evolves as the child grows. Using the proposed methodology, for the first time in the literature we characterize temporal changes in correlations among bacterial families during a baby's first year after birth. For example, we discover for the first time that *Lachnospiraceae* and *Ruminococcaceae*, which play important roles in degrading complex polysaccharides into short-chain fatty acids, interact with several families during a baby's first year of birth. Similarly, *Bifidobacteriaceae*, *Bacteroidiceae*, *Lachnospiraceae*, and *Clostridiaceae*, are all negatively correlated with *Enterobacteriaceae*, a family that contains pathogens such as *Salmonella*, *Shigella* and *Escherichia coli*. Statistical properties of the proposed methodology are evaluated analytically and by using synthetic data.

Introduction

In any given ecosystem, members maintain complex interactions among themselves for the growth and stability of the whole community. This is also true for microbial ecosystems. Some bacteria display mutualism where they help or benefit from each other, commensalism where one benefits while the others are not affected, parasitism or mutual exclusion where one may benefit at the expense of another[1]. One needs to understand perturbations or changes in the interactions among microbiota within and between ecosystem(s) because they can potentially impact various health outcomes, such as obesity[2], inflammatory bowel diseases[3], HIV[4], and so on. To make such assessments, there is an urgent need for statistical and computational methods to characterize interactions among microbiota.

Although there exists considerable literature on methods for performing differential abundance analysis and analysis of diversity[5,6], methods for describing interactions among microbiota are not as well developed. As the first step to characterize interactions, one typically is interested in describing the correlation coefficient between a pair of microbes.

16S ribosomal RNA (rRNA) and other high-throughput sequencing techniques enable profiling of microbial communities, revealing the abundances and phylogeny of microbial populations across diverse ecosystems. The observed counts of taxa are constrained by an arbitrary sequencing depth (library size), which is due to a fixed upper bound on the number of reads per sequencing instrument. Thus, the observed counts are an unknown fraction of the underlying true abundance of taxa in a unit volume of the ecosystem[6,7]. Consequently, they are relative abundances and hence are compositional data[7–13]. It is well-recognized in the literature that due to compositionality of the observed microbiome data, standard methods such as Pearson, or Spearman correlation coefficients for these data are theoretically invalid which results in misinterpretations of the data[9,14–17]. The effects of compositionality on these methods are pronounced when the number of taxa is limited, such as microbiome data at the phylum level, and when the microbial diversity is small[14]. Additionally, sequencing efficiencies are not necessarily the same for each taxon, which leads to a taxon-specific bias when some taxa are preferentially measured over others during the sequencing experimental workflow[18]. A concrete example of this bias can be seen by the differential sequencing efficiencies between gram-negative and gram-positive bacteria. Gram-positive bacteria have strong cell walls, which makes them harder to extract than gram-negative bacteria in the data preparation step. Thus, gram-positive bacteria may be underrepresented in the observed abundances. We illustrate the problems of using standard concepts of correlation coefficients with the help of a toy example shown in Fig. 1. A total of 100 taxa were generated from a Negative-Binomial distribution. For simplicity of visualization, scatter plots corresponding to 5 taxa are shown in the figure. The data were generated so that the underlying true abundances for taxa T1 and T3 are perfectly linearly correlated, whereas taxa T2 and T4 are parabolic in their relationship, and the remaining pairs are uncorrelated (Fig. 1a). These correlations were completely distorted in the observed counts (Fig. 1b) due to differential sampling fractions[6,7] (across samples) and sequencing efficiencies (across taxa). Without

properly addressing these two sources of biases, the standard concepts of correlation coefficients between a pair of taxa, such as the Pearson and the Spearman correlation coefficients, are erroneously positive (Fig. 1c) with very significant p-values. The distribution of p-values is far from the diagonal line corresponding to uniform distribution (Supplementary Fig. 1a). This phenomenon is purely driven by the artifacts of variations in sampling fractions and sequencing efficiencies.

A second challenge is that the next-generation sequencing (NGS) technologies yield massive high-dimensional data, where the number of microbiota d often far exceeds the sample size n. Since d often exceeds n, statistically the correlation matrix of the microbiota is singular (positive semi-definite matrix)[19], unless a condition of sparsity is imposed. In simple terms, sparsity means that many pairwise correlations are zero. Thus, compositionality aside, due to the high dimensionality, it is well-known that the sample Pearson and Spearman correlation matrices provide biased estimates of the correlation matrix[20]. A host of approaches have been proposed to estimate the covariance matrices in high dimensions[21–29]. However, since the observed high dimensional microbiome data contain a large number of zeros and are compositional, problem of estimating the correlation matrices in the present setting further complicates. It is well-known that microbiome abundance table could contain as many as 90% zero entries[6,30], therefore it is reasonable to assume that most taxa are not correlated or at best weakly correlated with each other. Thus, the sparsity assumption noted above is reasonable in the present context.

Thirdly, since microbiota have complex inter-dependencies between each other, not all inter-relationships are expected to be linear. Consider, for example, the relationship between *Ruminococcaceae* and *Staphylococcaceae* depicted in Supplementary Fig. 2, which is based on data obtained from the Norwegian Microbiome study (NoMIC)[31,32]. While, linear fit seems reasonable, the relationship appears to be very nonlinear. In more complex settings, nonlinear correlations among taxa within an ecosystem or across systems (e.g., gut and oral environments) are ubiquitous[33]. Such nonlinear interactions among microbiota are known to influence microbial composition and host health[34,35]. However, standard concepts of correlation such as the Pearson or Spearman correlation coefficients are designed for quantifying linear relationships and not suitable for quantifying nonlinear relationships. Recent methods such as Sparse Correlations for Compositional data (SparCC)[14], proportionality[15], and Sparse Inverse Covariance Estimation for Ecological Association Inference (SPIEC-EASI)[16] were developed for computing compositional correlations. However, they too are designed for quantifying linear relationships and may not be suitable for nonlinear relationships. For instance, in the example shown in Fig. 1, the relationship between T2 and T4 is parabolic, and the correlation coefficient provided by SparCC is zero (Fig. 1d), suggesting no linear relationship. Researchers often misconstrue that two taxa are independent if correlation coefficients such as the above are zero. It is also important to note that although SparCC is designed for quantifying linear relationships in compositional data, it may falsely declare linear relationship when two taxa are truly uncorrelated in the ecosystem. For example, taxa T3 and T4 are uncorrelated in the ecosystem (Fig. 1), and yet, SparCC resulted in a significant non-zero correlation coefficient (Fig. 1d).

To quantify nonlinear correlations between a pair of variables, the concept of distance correlation was introduced in the literature for the Euclidean space data[36]. The difference between distance correlation and Pearson/Spearman correlations is illustrated in Supplementary Fig. 3. All three measures (Pearson, Spearman, and distance correlations) detected the relationship between x and y with a correlation coefficient of 1 when the two random variables are linearly correlated. However, when x and y are nonlinearly related, e.g., $y = (x-5)^2$, the Pearson and Spearman correlations were both 0, whereas the distance correlation was ~ 0.5. It is important to note that a value of zero distance correlation implies statistical independence between two variables whereas a zero Pearson or Spearman or SparCC correlation coefficients only implies lack of linear relationship.

Lastly, correlation coefficient methods for compositional data such as SparCC, proportionality[15], and SPIEC-EASI[16] are scale invariant for computing correlations between taxa within a single ecosystem. Note that a function f is scale invariant if $f(ax) = f(x)$ for every $x \in \mathbb{S}^d$ and for every $a > 0$, where $\mathbb{S}^d = \{\vec{x} = [x_1, \ldots, x_d] | x_i > 0, i = 1, \ldots, d; \sum_{i=1}^{d} x_i = \kappa\}$ is a simplex[37]. However, these methods are not scale invariant when computing correlations between taxa across two or more ecosystems.

Motivated by the above limitations of the existing methods, in this article we introduce a new methodology called Sparse Estimation of Correlations among Microbiomes (SECOM) which provides two measures of correlations between a pair of taxa, one for linear relationships and the other for nonlinear relationships using the concept of distance correlations. As can be seen in Fig. 1e, SECOM is useful for quantifying linear as well as nonlinear relationships. In the Methods section of the paper we provide guidance on how to use these measures when describing dependencies between a pair of taxa. The proposed methodology is statistically rigorous and accounts for compositionality and differential sequencing efficiencies by explicitly estimating library-specific sampling fraction[6,7] and taxon-specific sequencing efficiency[38,39]. The resulting observed counts, corrected for two sources of biases will be called "bias corrected" counts which can be used for drawing inferences on the abundance of taxa in a unit volume of an ecosystem[6,7]. Furthermore, as can be seen from the p-value distribution provided in Supplementary Fig. 1b, unlike the existing methods, SECOM does not suffer from inflated false correlations between taxa.

The performance of SECOM is evaluated using a variety of simulations studies. The results are described in the Results section of this paper. The statistical details are provided in the Methods section of the paper. In addition to results based on synthetic data, we also illustrate SECOM using the infant gut microbiome data from the Norwegian Microbiome study (NoMIC). All theoretical proofs are provided in the Supplementary Notes.

Results

Performance of SECOM using simulated data

Using simulated data, in this section we compare the performance of SECOM with some existing methods, namely, proportionality[15], SparCC[14], SPIEC-EASI[16], and standard Pearson correlation coefficient. All details regarding the simulation study design are provided in the Supplementary Notes.

We begin by comparing CPU run times of various methods for computing correlation matrices using a simulated data set consisting of 200 taxa and 100 samples. All comparisons are made using RStudio with 1 CPU, x86_64-w64-mingw32/x64 (64-bit) and Windows 10 x64 (build 18363). Results are summarized in Supplementary Table 1. On average, SPIEC-EASI using either network inference method (MB or GL) took nearly hundred times more CPU time than other approaches, while SECOM was competitive with other compositional methods, such as SparCC and Proportionality.

Next we compare the performance of SECOM with other methods for estimating the correlation matrix. We use the following average relative norm loss as the criterion for evaluating the performance of a method:

$$\mathbb{E}\left\{\frac{\|\hat{R}-R^0\|_*}{\|R^0\|_*}\right\}.$$

We use the two commonly used matrix norms $\|\|_*$, namely, Frobenius norm ($\|A\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$) and spectral norm ($\|A\|_\infty = \lambda_{\max}(A)$, which is the largest singular value of A). Additionally, we also evaluated the performance of the various methods in their ability to discover true non-zero correlations, defined as the true positive rate (TPR), and falsely declaring non-zero correlations when the correlations are truly zero, defined as false positive rate (FPR). Thus, estimated TPR and FPR are as defined below:

$$\text{TPR} = \frac{\#\{(l,m) : \hat{\rho}_{lm} \neq 0, \rho^0_{lm} \neq 0\}}{\#\{(l,m) : \rho^0_{lm} \neq 0\}},$$
$$\text{FPR} = \frac{\#\{(l,m) : \hat{\rho}_{lm} \neq 0, \rho^0_{lm} = 0\}}{\#\{(l,m) : \rho^0_{lm} = 0\},}$$

where $R^0 = [\rho^0_{lm}]_{l,m=1,\ldots,d}, \hat{R} = [\hat{\rho}_{lm}]_{l,m=1,\ldots,d}$.

Simulation studies are based based on three different scenarios of ecosystems, namely, (1) an ecosystem where some taxa are linearly correlated and the rest uncorrelated, (2) an ecosystem where some taxa are nonlinearly correlated and the rest are uncorrelated, and (3) two ecosystems where some taxa are linearly correlated between ecosystems and the rest (either within each ecosystem or between ecosystems) are uncorrelated. Sparse correlation matrix obtained from SECOM is either by a data-driven thresholding approach (denote it as "SECOM (Thresholding)") or by p-value filtering (denote it as "SECOM (Filtering)").

Linear correlations. Fig. 2 summarizes the results using different approaches to estimate the correlation matrix under different sample size/taxon number/dispersion combinations ($n/d/\alpha$). As expected, under both norms, all compositional methods had uniformly smaller average relative norm loss than the standard Pearson correlation coefficient (Fig. 2a). For example, with 100 samples, 200 taxa, and the dispersion parameter equals to 0.5, the standard Pearson correlation coefficient had approximately 9 times higher average loss under both norms in comparison to SECOM (Frobenius: 9 vs. 1; Spectral: 45 vs. 5). The SparCC method had the second largest average loss especially when there is greater variability in the underlying true abundances (overdispersion parameter $\alpha = 2$, Fig. 2a). Both versions of SECOM, namely, Thresholding and Filtering, together with proportionality and both SPIEC-EASI methods outperformed other methods in terms of average loss (Fig. 2a). In addition to the estimation accuracy, SECOM (Thresholding and Filtering) and SPIEC-EASI (MB and GL) also performed well in recovering the sparsity pattern (Fig. 2b). They not only outperformed other methods substantially in controlling FPR, but they competed well in terms of TPR. However, as mentioned above, SPIEC-EASI is substantially more computationally intensive than SECOM. Under this simulation setting, SPIEC-EASI required a CPU time of 76.15 hours in comparison to 1.2 CPU hours for SECOM. Both SparCC and standard Pearson correlation coefficients achieved high TPR in this simulation study but were subject to very high FPR as compared to SECOM and SPIEC-EASI. Note that the FPR of standard Pearson correlation can be as large as 80%, which could make most discoveries meaningless and even misleading. Although proportionality performed well in terms of FPR, it has very small TPR when there are larger variations in taxa abundance. For example, when the underlying true dispersion parameter increased to 2, the TPR of proportionality method dramatically reduced to almost 0. It is also worth noting that for a fixed dispersion parameter, as the sample size n and the number of taxa d increased, there is an increase in TPR together with a decrease in FPR for both SECOM methods. This is consistent with the concentration results described in Theorem 0.1.

Nonlinear correlations. We also performed a simulation study comparing the performance of various methods in identifying nonlinear relationships. The settings were similar to the simulation study conducted for linear correlations described above, except that each taxon was set to be quadratically correlated with its adjacent taxon among the first 50 taxa. Since, with the exception of SECOM, none of the existing methods are designed to measure nonlinear relationships, we only compared these methods in their ability to recover sparsity. We implemented SECOM with p-value filtering strategy (i.e., SECOM (Filtering)) but did not implement the data-driven thresholding strategy (i.e., SECOM (Thresholding)) because it involves a step of data splitting to divide data into training and test sets, which requires much larger sample size when it comes to the detection nonlinear relationships. The sparsity strategies for SparCC and proportionality, and the tuning parameters for SPIEC-EASI used in this simulation set-up are same as those used for linear case mentioned earlier. As expected, none of the competing methods identify nonlinear relationship (Fig. 3). They all have TPRs approximately 0. Although standard Pearson correlation coefficient achieves a TPR at around 50%, but its FPR was sometimes even higher, leading to misleading discoveries when detecting complex microbial inter-relationships. With the adoption of distance correlation concept, SECOM successfully achieved almost perfect TPR (100%), and also had significantly smaller FPR than existing methods.

Multiple ecosystems. In addition to quantifying correlations between taxa in a single homogeneous ecosystem, we also compared SECOM with existing methods in quantifying microbial correlations across two ecosystems. Specifically, a simulation study was conducted by generating count tables from two ecosystems with different sampling fractions and sequencing efficiencies in each simulation run. The first 50 taxa in the first ecosystem were linearly correlated with the corresponding taxa in the second ecosystem. Rest of the taxa are uncorrelated within or between ecosystems. Details of the study design are provided in the Supplementary Notes. For all the methods, we used the same methodology as described in the other simulation studies mentioned above. As seen in Fig. 4, SECOM methods outperformed all competitors in terms of both estimation accuracy (the smallest average relative Frobenius/Spectral norm loss) and sparsity recovery (the smallest FPR and the highest TPR). In general, SECOM (Thresholding) had the most precise correlation estimate, the smallest FPR, and comparable TPR as SECOM (Filtering). Both SECOM methods achieve almost perfect TPR ($\sim 100\%$) while SECOM (Filtering) performed slightly better. On the contrary, standard Pearson correlation coefficients together with existing compositional techniques either suffered from loss of TPR, or inflated FPR, making them unsuitable for detecting microbial inter-relationships across ecosystems.

As a summary of simulation studies, SECOM performed consistently well regardless of whether correlations are linear or nonlinear. The thresholding version of SECOM is entirely data-driven and is theoretically proved to be a consistent estimator of the true correlation matrix in the Frobenius norm (Theorem 0.3 in the Methods section). However, since the thresholding method requires splitting data into training and test sets, it generally needs a larger sample size especially when it comes to the detection of nonlinear relationships. On the other hand, the filtering version of SECOM screens the estimates of correlation coefficients based on their p-values. Correlation coefficients with statistically non-significant p-values are discarded to achieve sparsity. The filtering method requires smaller sample size than thresholding method as there is no data splitting step. However, the determination of significance level is heuristic since controlling the type I error is not the primary purpose of SECOM. For example, setting the significance level $\alpha = 0.05$ does not mean the overall type I error rate (or family-wise error rate (FWER), or false discovery rate (FDR) if the multiple comparison correction is implemented) is controlled at 5%, especially for the detection of nonlinear relationships. It should simply be viewed as a filtering mechanism to achieve sparsity. For simplicity of exposition, we only implemented SECOM (Filtering) in the following Illustration section.

Illustration: Norwegian infant gut microbiome data

We illustrate SECOM using the data obtained from the Norwegian Microbiome study (NoMIC)[31,32]. The prospective NoMIC study was established to study the development of infant gut microbiota and its relation to health outcomes. Mothers were recruited from the maternity ward at a county hospital in south-eastern Norway from 2002 to 2005. Stool samples were obtained at days 30, 120 and 365 after birth. Samples were retrieved from the mothers and kept frozen at -20°C until DNA was extracted by an automated procedure and amplified by PCR reactions using 16S rRNA specific primers. The V4 region of the 16S rRNA gene was then sequenced using Illumina HiSeq. Details on the study population have been reported previously[31,32]. Written informed consent was obtained from the mothers prior to enrolment. The study was approved by the Norwegian Data Inspectorate, and the Norwegian Regional Committee for Medical and Health Research Ethics. In this illustration we restricted to children who were vaginally delivered with no perinatal antibiotics exposure and all exclusively breastfed during the first four months of life. We chose to study the correlations at 30, 120, and 365 days as they represent different feeding patterns, which resulted in 46, 44, and 34 sequenced samples, respectively. As an infant grows from infancy to toddler stage, their environment changes dramatically due to changes in diet (from breastfeeding to solid foods), external contacts, environment, and activities. These changes necessarily impact their gut microbiota and likely contributes to normal growth and development of a baby[40,41]. We focused on analyzing data at the family level. The prevalence for each family is provided in Supplementary Table 2.

Bar graphs of temporal changes in the bias corrected counts and relative abundances of top ten prominent families are

provided in Fig. 5. Similar plots at the phylum level are provided in the Supplementary Fig. 4. Since all babies are breastfed during the first 30 days after birth and are in constant maternal contact, consistent with literature, we found greater relative abundance of the phylum Actinobacteria, specifically the family *Bifidobacteriaceae* (55%), along with families of Firmicutes, such as *Clostridiaceae* (10%), *Lachnospiraceae* (1%), *Ruminococcaceae* (9%), and opportunistic *Staphylococcaceae* (1%). Since these babies were born vaginally, we also see some *Bacteroidaceae* (6%) during the first 30 days. We also found pro-inflammatory bacterial families *Enterobacteriaceae* (9%) and *Pasteurellaceae* (2%), in the phylum Proterobacteria in the 30-day old babies. As the baby's environment and activities begin to change, in particular transitioning from breastfeeding to other foods, we begin to notice changes in the microbial compositions in our cohort of babies. As one would expect the relative abundance of *Bifidobacteriaceae*, which is largely due to breastfeeding, temporally decreases by the end of the first year (55%, 35%, 5% at Day 30, 120, 365, respectively). We also notice changes in the relative abundance of *Clostridiaceae* (10%, 30%, 3% at Day 30, 120, 365, respectively). The relative abundance of *Enterobacteriaceae* was generally constant throughout the first year (9%, 9%, 6% at Day 30, 120, 365, respectively). As babies begin to crawl, touch various objects and are in contact with other people or environment, one would expect several new families to colonize the gut by Day 365. For example, we see the arrival of *Moraxellaceae* (1%) in the gut at Day 365 and numerous other families which are grouped as "Others" (68%) .

To the best of our knowledge, this is the first paper to characterize correlations among different families of bacteria during the first year of development of a baby. The correlations (linear and nonlinear) among the top 10 prevalent families are summarized in Fig. 6. To deal with sparsity, we filtered all correlations that were below 0.1 and p-value exceeding 0.1 by setting them to zero. If the estimated linear as well as distance correlations are non-zero, then it suggests a potential linear or nonlinear association between the two microbes. If the estimated linear correlation is zero but the distance correlation is different from zero then it suggests a nonlinear relationship between the pair of taxa. However, due to power, sometimes a linear correlation is significantly different from zero even though the nonlinear correlation is not significantly different from zero.

During the first 30 days of birth, important families of bacteria that promote baby's health begin to colonize the gut with little or no interactions among themselves. However, we see significant negative linear correlations between *Enterobacteriaceae* and *Bifidobacteriaceae* (linear correlation -0.44, distance correlation 0.44) and between *Enterobacteriaceae* and *Bacteroidiceae* (linear correlation -0.61, distance correlation 0.61). As noted above, Day 120 is a critical transition age when the gut microbial diversity increases as new taxa arrive in the gut. Consequently, we hypothesize that inter-relationships among the gut microbiota changes from how they were at Day 30. We discovered strong negative linear relationships between the pathogenic family *Enterobacteriaceae* several important commensal families such as *Lachnospiraceae* (linear correlation -0.89, distance correlation 0.92), *Clostridiaceae* (linear correlation -0.75, distance correlation 0.73) and *Ruminococcaea* (linear correlation 0, distance correlation 0.51). We also saw negative correlation between *Bifidobacteriaceae* and *Clostridiaceae* (linear correlation -0.49, distance correlation 0). By Day 365, an infant's physical activity, diet and environment would substantially change and hence many new families start colonizing the gut and developing complex interactions among them. Again, we see *Lachnospiraceae* increase interactions with other microbes with age. At Day 365, as the (bias corrected) abundance of phylum Actinobacteria decreases (Supplementary Fig. 4) in particular as the abundance of *Bifidobacteriaceae* decreases, *Lachnospiraceae* begins to increase (Fig. 5a) and the two families are linearly negatively correlated with a correlation coefficient of -0.45 (distance correlation 0.5). *Lachnospiraceae* is also negatively correlated with *Enterobacteriaceae* with a linear correlation of -0.37 (distance correlation 0.42), and with opportunistic family *Staphylococcaceae*, with a linear correlation of -0.74 (distance correlation 0.78). *Lachnospiraceae* is also nonlinearly correlated with another important gut microbiota *Ruminococcaceae*, with a distance correlation of 0.42. Similar to *Lachnospiraceae*, the *Ruminococcaceae* family is also found to be negatively correlated with the pathogen *Enterobacteriaceae* with a linear correlation of -0.42 (distance correlation 0.45) and also negative correlated with opportunistic family *Staphylococcaceae* with a linear correlation coefficient of -0.56 (distance correlation coefficient of 0.62). Not surprisingly, *Ruminococcaceae* family is positively linearly correlated with *Bacteroidaceae* with a correlation of 0.34 (distance correlation 0.37). We see nonlinear relationship between *Ruminococcaceae* and *Bifidobacteriaceae* with a distance correlation 0.36. We also find a nonlinear relationship between *Bifidobacteriaceae* and opportunistic family *Staphylococcaceae* with a distance correlation 0.56. A nonlinear relationship was seen between *Bacteroidaceae* and *Pasteurellaceae* with a distance correlation of 0.56. A positive correlation between *Enterobacteriaceae* and *Clostridiaceae* of 0.36 was seen (distance correlation 0.39). Thus, in most cases we notice that the distance and linear correaltion seem to track each other. However, there are instances where the relationships appear to be nonlinear with a zero linear correlations.

In summary, we find that starting Day 120, *Lachnospiraceae* and *Ruminococcaceae* to be interactive with several families during the first year of a baby's growth and development. These two families play an important role in degrading complex polysaccharides to short-chain fatty acids such as acetic, butyric, and propionic acid, which have numerous biochemical functions[42]. Thus, it is not surprising that during the development of a baby, these families interact with many other families of bacteria to achieve a balance in the gut ecology, especially as the baby begins to ingest more complex foods beyond breast milk. Not surprisingly, during the early days of gut colonization (Day 30 and Day 120), important families that promote infants'

health, such as *Bifidobacteriaceae*, *Bacteroidiceae*, *Lachnospiraceae*, and *Clostridiaceae*, are all negatively correlated with *Enterobacteriaceae*, a family of bacteria that contains many familiar pathogens, such as *Salmonella*, *Shigella* and *Escherichia coli*.

It is important to note that microbial ecology is very complex and dynamic and hence it is not reasonable to expect very high correlation coefficients. This is because, a family of bacteria consists of numerous genera and many species within a genus. Thus, relationships between a pair of families could be affected by complex inter-relationships between species or even strains of bacteria. The description provided in this paper should be viewed as proof of concept of the methodology. Should a researcher be interested in drilling down the correlations at a lower level of the phylogenetic tree, then larger sample sizes would be necessary.

Discussion

Humans have co-evolved with microorganisms since the beginning of time. A symbiosis exists whereby functions performed by the human microbiome did not need to be established by the human host. However, modern lifestyle has severely depleted the commensal microbiomes in the human host, whereby many functions humans rely on, have been lost. This conveys a risk of lower well-being and disease. However, it also offers an opportunity for prevention and treatment of diseases if we were to replenish the gut microbiome back to an optimal state. This prospect offers great promise, yet has to be based on knowledge of what constitutes an optimal gut microbial ecosystem, of which we yet possess limited knowledge. Moreover, replenishing the ecosystem is not straightforward as introducing new microbes will affect microbes already present, thus we need to learn more about how microbes and microbial groups are correlated. Additionally, correlations are likely different in an ecosystem of ancestral microbes in contrast to an ecosystem disrupted by medical interventions. Thus, we chose to study correlations in a selected subset of infants who had not been subjected to medical interventions such as cesarean delivery and perinatal antibiotics, know to profoundly disrupt the microbiome for a prolonged period and with long term effects on health.

Understanding interactions among microbes is a challenging scientific problem. It is further complicated by the statistical challenges due to the compositional structure in the data. Traditional methods to describe correlations, such as Pearson and Spearman correlation coefficients are not valid. Further, as demonstrated in this paper, relationships between a pair of microbes is not always linear as there exist complex interactions among them. Unfortunately, Pearson and Spearman correlation coefficients are measures of linear associations and may not be suitable to describe nonlinear relationships. This paper introduces a measure of linear as well as nonlinear relationship between a pair of microbes. The resulting methodology is called Sparse Estimation of Correlations for Microbiomes (SECOM). The new methodology overcomes the various challenges experienced with microbiome data.

Application of SECOM to an infant gut microbiome data set obtained from the NoMIC study, lets us discover interesting temporal patterns in correlations between a pair of taxa. Correlations among various microbes evolves with the development of a baby. During the early days of life (30 days from birth), new microbes are continuously colonizing the gut. Four months after birth is a critical window in an infant's growth and development when many babies begin are weaned from breast milk to solid foods. In response, the gut at this age undergoes microbial changes and accordingly the correlations among microbiota evolve with age. By the time an infant is about one year old, the infant gut matures with the inclusion of many new microbial families. Accordingly, the gut undergoes further transformation and many new relationships among bacteria are formed. For the first time in the literature, we discover linear and nonlinear relationships among various families. By investigating the group of babies who were vaginally born, breastfed for the first four months of life and not exposed to antibiotics during this time, we develop a baseline understanding of naturally occurring correlations among various families. In the future, we would be interested in understanding how these relationships are modified when there has been clinical interventions such as C-section and the use of antibiotics, which are becoming popular in the modern societies around the world. Subsequently, we will also be interested in investigating the effects of these associations on various health outcomes among infants. A limitation of the present study is that due to the small sample size, we may have missed some significant correlations.

Methods

Description of some existing methods

Proportionality The ϕ statistic was proposed to describe the strength of proportionality between two variables, which is closely related but not identical to correlation coefficient. Instead, it is a "goodness-of-fit to proportionality" statistic to assess the extent to which a pair of random variables are proportional[15]. $\phi = 0$ indicates two random variables are exactly proportional (e.g. $y = ax$); on the other hand, $\phi \neq 0$ as long as two random variables are not exactly proportional, even if they are perfectly linearly correlated (e.g. $y = ax + b$). Additionally, unlike the standard concept of correlation coefficients, it is hard to interpret ϕ values as it lacks a scale. For instance, $\phi = 0.1$ does not imply how strong two random variables are related to each other without considering all ϕ statistics. If all ϕ statistics range from 0 to 1, $\phi = 0.1$ could indicate a strong proportionality; while if

all ϕ statistics range from 0 to 0.1, $\phi = 0.1$ could otherwise represent a weak proportionality. Also, ϕ statistics is not symmetric ($\phi(x,y) \neq \phi(y,x)$), making it inappropriate to serve as a dissimilarity measure for downstream analyses such as ordination or clustering analysis. Lastly, there is no clear criterion to produce a sparse matrix based on ϕ statistics.

Sparse Correlations for Composition Data (SparCC)[14] It is designed to estimate the correlation coefficients for log-transformed abundances while addressing bias due to compositionality in the data. It assumes that (1) the number of taxa is large, and (2) the underlying true correlation matrix is sparse. However, similar to proportionality method, it is not a statistically rigorous approach to obtain a sparse correlation matrix for SparCC. An arbitrary cutoff of $r = 0.3$ was suggested in[14] to declare if a pair of taxa are correlated or not.

Sparse Inverse Covariance Estimation for Ecological Association Inference (SPIEC-EASI[16]) It is a measure of association that exploits the concept of conditional independence and graphical models. Two taxa are conditionally independent if given the abundances of all other taxa, neither taxon provides additional information about the abundance of the other. Otherwise, these taxa are not conditionally independent and there is a linear relationship between them as the method is essentially estimating the inverse covariance matrix (precision matrix). While conceptually it is a useful measure, as one would expect, SPIEC-EASI is computationally is very intensive compared to the above existing methods.

SECOM model set-up

Term	Definition
i	Sample, $i = 1,\ldots,n$.
j	Taxon, $j = 1,\ldots,d$.
A_{ij}	The absolute abundance in the ecosystem for taxon j in sample i.
O_{ij}	The observed abundance for taxon j in sample i.
S_i	Sample-specific sampling fraction.
C_j	Taxon-specific sampling efficiency.
E_{ij}	Random error for taxon j in sample i.
a_{ij}	$\log A_{ij}$.
o_{ij}	$\log O_{ij}$.
s_i	Sample-specific sampling fraction in log scale.
c_j	Taxon-specific sequencing efficiency in log scale.
e_{ij}	Random error for taxon j in sample i in log scale.

Table 1. Definitions of key notations

The absolute abundances A_{ij} in the ecosystem are non-negative. Let $\mathscr{T}$ be the set of all possible configurations of presence or absence of a taxon, then

$$\mathscr{T} = \begin{cases} (1,1,\ldots,1) \\ (0,1,\ldots,1),\ldots,(1,1,\ldots,0) \\ \vdots \\ (0,0,\ldots,1),\ldots,(1,0,\ldots,0) \end{cases}$$

where 0 and 1 correspond to the cases where a taxon is missing (zero) or present in a sample, respectively. We represent each event in $\mathscr{T}$ by T and it is indexed by $t = 1,\ldots,2^d - 1$. For example, T_1 corresponds to the case where all taxa are present in a sample, and T_{2^d-1} is the case where only the first taxon is present, and all remaining taxa are zeros.

We further assume that for each sample, the pattern of taxon presence is generated by independent r.v. $M_{ij}, 1 \leq i \leq n, 1 \leq j \leq d$ such that

$$P(M_i = T_t) = \delta_t^i, 0 \leq \delta_t^i \leq 1,$$

where $M_i = (M_{i1},\ldots,M_{id})^T$.

We propose that the observed abundance is generated by the following multiplicative model:

$$O_{ij} = S_i C_j A_{ij} E_{ij}. \tag{1}$$

For A_{ij} with $M_{ij} = 1$, i.e., for the non-zero absolute abundance, perform the log transformation, the above multiplicative model can be transformed into an additive model,

$$o_{ij} = s_i + c_j + a_{ij} + e_{ij}. \tag{2}$$

Assume that conditional on $M_i = 1$, the absolute abundance has finite second moment:

$$E(a_{ij}) = \alpha_j, Var(a_{ij}) = \sigma^0_{jj}. \tag{3}$$

Additionally, we assume that:

s_i is a sample-specific constant,

c_j is a taxon-specific constant,

$$e_{ij} \sim_{i.i.d.} F_e \text{ with } E(e_{ij}) = 0, Var(e_{ij}) = \sigma^2_e, \tag{4}$$

$$a_{ij} \perp\!\!\!\perp e_{ij}. \tag{5}$$

We define two concepts of correlation coefficients in this section. One quantifies the strength of linear relationship between a pair of taxa by modifying Pearson and Spearman correlation coefficients for sparse high-dimensional compositional data. The second quantifies the strength of nonlinear relationship between a pair of taxa by modifying distance correlation coefficient. We provide both measures.

Remark 1. Since distance correlation coefficient is a measure of nonlinear relationship between a pair of taxa, it does not have a sign and is always non-negative. On the other hand, the measure of linear correlation coefficient described below can take positive or negative values depending upon whether they are positively or negatively linearly correlated.

Remark 2. A distance correlation coefficient of zero between two taxa implies that the taxa are independent. It is well-known that if Pearson or Spearman correlation coefficients between two taxa is zero then it only suggests a lack of linear relationship but not independence.

Remark 3. Our proposed SECOM methodology outputs both linear as well as distance correlation coefficients between pairs of taxa. If the relationship between a pair is nonlinear then we expect the distance correlation between the two taxa to be nonzero whereas the linear correlation could be zero. On the other hand if the relationship is linear then the linear correlation coefficient is expected to be nonzero and sign of the coefficient is positive or negative depending upon whether the two taxa are positively or negatively associated. When the linear correlation coefficient is nonzero then we expect the distance correlation coefficient also to be nonzero. This is because linear correlation coefficient is a special case of nonlinear correlation coefficient. In such cases, we recommend the user to use the value provided by the linear correlation coefficient. However, there may be situations where the estimated linear correlation is nonzero but the distance correlation is zero. This can happen due sample size and power issues and the sparsity constraints. In such cases again we recommend the researcher to use linear correlation coefficient value.

Estimation of linear correlation coefficients

In this section we develop methodology for estimating linear correlation coefficient between the abundances of pairs of taxa l and m, $1 \le l, m \le d$, while noting that for each subject i, the abundances a_{il} and a_{im} are not observable. The only observable values are the observed counts, e.g. OTUs, o_{il} and o_{im}. According to our hypothesized model, the actual abundance a_{ij} of the jth taxon in a unit volume of the specimen from the ith subject, in expectation, is a function of observed counts o_{ij}, and unknown nuisance parameters s_i and c_j. Therefore, to obtain an approximate unbiased estimator of a_{ij} using the observable o_{ij}, we need to estimate and nuisance parameters s_i and c_j and eliminate the bias introduced by them. Once that is done, we can define a Pearson or Spearman correlations using the modified observed counts o_{ij} to derived bias-corrected estimates of the true correlation coefficient between the abundance of a pair of taxa. In the following we now describe the methodology to eliminate the nuisance parameters s_i and c_j.

Denote the covariance matrix of a_{ij} as $\Sigma^0 = [\sigma^0_{lm}]_{l,m=1,\ldots,d}$, and the Pearson correlation coefficient matrix as $R^0 = [\rho^0_{lm}]_{l,m=1,\ldots,d}$. We make the following sparsity assumption regarding the covariance matrix.

Assumption 0.1 (Approximately sparse covariance matrices)**.**

$$\mathscr{U}(K) = \{\Sigma : \sigma_{jj} < K, \frac{1}{d^2} \sum_{l \neq m} \sigma_{lm} = o(1)\}.$$

Two examples of matrices which satisfy this condition are (1) a d-diagonal matrix, and (2) the AR(1) covariance matrix.

From (2), (3), (4), and (5), by centering the data suitably, we eliminate the effect of taxon-specific sampling efficiency c_j as follows:

$$o_{ij} - \bar{o}_{\cdot j} = (s_i - \bar{s}_{\cdot}) + (a_{ij} - \bar{a}_{\cdot j}) + (e_{ij} - \bar{e}_{\cdot j}) = \textcircled{1} + \textcircled{2} + \textcircled{3},$$

where, for the jth taxon and $i = 1, 2, .., n$, $\bar{s}_{\cdot} = \frac{1}{n}\sum_{i=1}^{n} s_i$, $\bar{a}_{\cdot j} = \frac{1}{n}\sum_{i=1}^{n} a_{ij}$, $\bar{e}_{\cdot j} = \frac{1}{n}\sum_{i=1}^{n} e_{ij}$. Note that:

(1) We call $\textcircled{1}$ the "sampling fraction difference", which is a sample-specific bias term,

(2) For $\textcircled{2}$: $E(a_{ij} - \bar{a}_{\cdot j}) = 0, Var(a_{ij} - \bar{a}_{\cdot j}) = (1 - \frac{1}{n})\sigma^0_{jj}$,

(3) For $\textcircled{3}$: $E(e_{ij} - \bar{e}_{\cdot j}) = 0, Var(e_{ij} - \bar{e}_{\cdot j}) = (1 - \frac{1}{n})\sigma^2_e$.

Define the set of non-zero taxa in sample i by $d(i) = \{j : M_{ij} = 1, 1 \le j \le d\}$. Then by on Assumption 0.1, we have

$$\frac{1}{|d(i)|}\sum_{j\in d(i)}\left\{(o_{ij} - \bar{o}_{\cdot j}) - \textcircled{1}\right\} = \frac{1}{|d(i)|}\sum_{j\in d(i)}\textcircled{2} + \frac{1}{|d(i)|}\sum_{j\in d(i)}\textcircled{3} \longrightarrow_p 0 \text{ as } |d(i)| \to \infty. \tag{6}$$

Therefore, a quick and simple estimator of the sampling fraction difference is the arithmetic mean of the difference between observed abundances, i.e. $\widehat{\textcircled{1}} = \widehat{s_i - \bar{s}_{\cdot}} = \frac{1}{|d(i)|}\sum_{j\in d(i)}(o_{ij} - \bar{o}_{\cdot j})$. Taking into account the variance of residuals $o_{ij} - \bar{o}_{\cdot j}$, a weighted average is proposed as follows:

$$\widehat{\textcircled{1}} = \widehat{s_i - \bar{s}_{\cdot}} = \sum_{j\in d(i)} w_j(o_{ij} - \bar{o}_{\cdot j}), \tag{7}$$

where $w_j = \frac{1}{Var(o_{ij} - \bar{o}_{\cdot j})} / \sum_{l\in d(i)} \frac{1}{Var(o_{il} - \bar{o}_{\cdot l})}$.

Denote the *centered absolute abundance* by $x_{ij} = a_{ij} - \bar{a}_{\cdot j}, 1 \le i \le n, 1 \le j \le d$. Then

$$Cov(x_{il}, x_{im}) = (1 - \frac{1}{n})\sigma^0_{lm}, \quad Var(x_{ij}) = (1 - \frac{1}{n})\sigma^0_{jj}, \quad Corr(x_{il}, x_{im}) = \rho^0_{lm}.$$

Similarly, define the *centered absolute abundance with noise* by

$$y_{ij} := (a_{ij} - \bar{a}_{\cdot j}) + (e_{ij} - \bar{e}_{\cdot j}) = x_{ij} + (e_{ij} - \bar{e}_{\cdot j}) = (o_{ij} - \bar{o}_{\cdot j}) - (s_i - \bar{s}_{\cdot}),$$

and denote $Cov(y_{il}, y_{im}) := \sigma_{lm}, Corr(y_{il}, y_{im}) := \rho_{lm}, \Sigma = [\sigma_{lm}]_{l,m=1,\ldots,d}, R = [\rho_{lm}]_{l,m=1,\ldots,d}$. Then the off diagonal and diagonal elements of Σ are, respectively,

$$\sigma_{lm} = (1 - \frac{1}{n})\sigma^0_{lm}, \quad \sigma_{ll} = (1 - \frac{1}{n})(\sigma^0_{ll} + \sigma^2_e).$$

We make the following assumption that noise to signal ratio is very small.

Assumption 0.2 (Small noise-to-signal ratio)**.**

$$\sigma^2_e \ll \sigma^0_{ll}.$$

Then we have:

$$\rho_{lm} = \frac{\sigma_{lm}}{\sqrt{\sigma_{ll}\sigma_{mm}}} = \frac{\sigma^0_{lm}}{\sqrt{(\sigma^0_{ll} + \sigma^2_e)(\sigma^0_{mm} + \sigma^2_e)}} \approx \frac{\sigma^0_{lm}}{\sqrt{\sigma^0_{ll}\sigma^0_{mm}}} = \rho^0_{lm},$$

which means the Pearson correlation coefficient among $y'_{ij}s$ provides a good approximation to the Pearson correlation coefficient among $x'_{ij}s$ as well as among $a'_{ij}s$. Since $y_{ij} = (o_{ij} - \bar{o}_{\cdot j}) - (s_i - \bar{s}_{\cdot})$. From (7), we estimate y_{ij} by $\hat{y}_{ij} = (o_{ij} - \bar{o}_{\cdot j}) - \widehat{s_i - \bar{s}_{\cdot}}$. Thus, from (6),

$$\hat{y}_{ij} \longrightarrow_p (o_{ij} - \bar{o}_{\cdot j}) - (s_i - \bar{s}_{\cdot}) = y_{ij}, \text{ as } |d(i)| \to \infty. \tag{8}$$

In the sequel, for all $1 \le l,m \le d$, denote $n(l) = \{i : M_{il} = 1, 1 \le i \le n\}$ and $n(l,m) = \{i : M_{il} = M_{im} = 1, 1 \le i \le n\}$, where $n(l)$ is the number of samples where l^{th} taxon is observed, and $n(l,m)$ represents the number of samples where both l^{th} and m^{th} are observed. For each $1 \le l,m \le d$, let

$$\hat{\mu}_l = \frac{1}{|n(l)|} \sum_{i \in n(l)} \hat{y}_{il},$$
$$\hat{\sigma}_{lm} = \frac{1}{|n(l,m)|} \sum_{i \in n(l,m)} (\hat{y}_{il} - \hat{\mu}_l)(\hat{y}_{im} - \hat{\mu}_m),$$
$$\Sigma_n = [\hat{\sigma}_{lm}]_{l,m=1\ldots,d},$$
$$R_n = [\hat{\rho}_{lm}]_{l,m=1\ldots,d} = \left[\frac{\hat{\sigma}_{lm}}{\sqrt{\hat{\sigma}_{ll}\hat{\sigma}_{mm}}}\right]_{l,m=1\ldots,d}.$$

To prove the consistency of the proposed covariance estimator, we make the following mild Assumption 0.3 that each taxon (Assumption 0.3(i)) or each pair of taxa (Assumption 0.3(ii)) is present with a nonzero probability (δ_{min}).

Assumption 0.3 (Minimal probability)**.** *Define*

$$C_i(l) = \{t : \textit{taxon } l \textit{ is present in } T_t \textit{ in } M_i, 1 \le t \le 2^d - 1\},$$
$$C_i(l,m) = \{t : \textit{taxa } l \textit{ and } m \textit{ are present in } T_t \textit{ in } M_i, 1 \le t \le 2^d - 1\}.$$

There exists a constant $\delta_{min} > 0$ such that for any $1 \le l,m \le d$,

$$(i)\frac{1}{n}\sum_{i=1}^{n}\sum_{t \in C_i(l)} \delta_t^i = \delta_l > \delta_{min}, \quad (ii)\frac{1}{n}\sum_{i=1}^{n}\sum_{t \in C_i(l,m)} \delta_t^i = \delta_{l,m} > \delta_{min}.$$

Theorem 0.1. *Suppose Σ^0 satisfies Assumption 0.1 and Assumption 0.3 is valid, then*

$$\left\|R_n - R\right\|_\infty = O_p\left(\sqrt{\frac{\log d}{n}}\right).$$

Thus, the empirical correlation matrix R_n based on $\hat{y}'_{ij}s$ consistently estimates the correlation matrix R of y_{ij} in the *Sup* norm. Under the Assumption 0.2, R is a reasonable approximation to the true correlation matrix R^0. The details of the proof are provided in the Supplementary Notes.

Instead of Pearson correlation coefficient, if the Spearman correlation coefficient is of interest, then we convert $x_{ij}, y_{ij}, \hat{y}_{ij}$ into ranks $r^0_{ij}, r_{ij}, \hat{r}_{ij}$. Clearly, for $r^0_{ij}, r_{ij}, \hat{r}_{ij} \in \{1, 2, \ldots, n\}$,

$$\hat{r}_{ij} \longrightarrow_p r_{ij}, \text{ as } |d(i)| \to \infty, \tag{9}$$
$$r_{ij} \approx r^0_{ij} \text{ as long as Assumption 0.2 holds.} \tag{10}$$

With a slight abuse of notations, denote $Corr(r_{il}, r_{im}) = s_{lm}$, $Corr(\hat{r}_{il}, \hat{r}_{im}) = \hat{s}_{lm}$, $S = [s_{lm}]_{l,m=1,\ldots,d}$, $S_n = [\hat{s}_{lm}]_{l,m=1,\ldots,d}$. Then we have the following result.

Theorem 0.2. *Suppose Σ^0 satisfies Assumption 0.1 and Assumption 0.3 is true, then:*

$$\left\|S_n - S\right\|_\infty = O_p\left(\sqrt{\frac{\log d}{n}}\right).$$

The proof of this result relies on the observation that standardized rank $\zeta_{il} = \frac{\hat{r}_{il} - E(\hat{r}_{il})}{\sqrt{Var(\hat{r}_{il})}}$ is naturally bounded, which allows the applicability of the Hoeffding's inequality. The details of the proof are deferred to the Supplementary Notes.

When computing correlations of high-throughput sequencing data, it is reasonable to assume that only a small fraction of features (genes/proteins/taxa etc.) are correlated with each other, i.e. the correlation matrix is sparse. Inspired by the Sparse Estimation of the Correlation matrix (SEC) approach[43], we estimate the sparse correlation matrix by solving

$$\hat{R} = \arg\min_R \frac{1}{2}\|R - R_n\|_F^2 + \lambda |R|_1, \text{ s.t. } R \succeq \varepsilon I, R_{jj} = 1, j = 1, \ldots, d. \tag{11}$$

where

(1) $\|\cdot\|$ is the Frobenius norm,

(2) $R \succeq \varepsilon I$ means that $R - \varepsilon I$ is semipositive definite,

(3) λ is the regularization parameter for the l_1 norm.

Denote the non-diagonal support of R as $A_0 = \{l,m : l \neq m, \rho_{lm} \neq 0\}$ and its cardinality as q. The convergence of $\hat{R}$ to R in Frobenius norm is established in the following theorem.

Theorem 0.3. *Suppose Σ^0 satisfies Assumption 0.1 and Assumption 0.3 is valid, and $\lambda = M\sqrt{\frac{\log d}{n}}$ for some constant M, we have the estimator defined in (11) such that*

$$\|\hat{R} - R\|_F^2 = O_p(q\frac{\log d}{n})$$

A sketch of the proof is provided in the Supplementary Notes.

From the computational point of view, the penalized likelihood problem stated in (11) can be replaced with the soft thresholding operator

$$\hat{R} = sign(R_n)(|R_n| - \lambda)_+, \tag{12}$$

where the tuning parameter λ is selected by cross-validation $\lambda = \min_\lambda \|\hat{R} - R_n\|_F$.[23]

Since the solution $\hat{R}$ may not be symmetric in general, the final estimate of $\hat{R}$ is obtained by forcing $\hat{\rho}_{ll} = 1$ and choosing $\hat{\rho}_{lm} = \hat{\rho}_{lm} I(\hat{\rho}_{lm} \leq \hat{\rho}_{ml}) + \hat{\rho}_{ml} I(\hat{\rho}_{lm} > \hat{\rho}_{ml})$. We denote the $\hat{R}$ obtained from the thresholding procedure as "SECOM (Thresholding)".

The sparse matrix $\hat{R}$ can also be obtained by thresholding on the basis of on p-values. Let p_{lm} be the corresponding p-value for $\hat{\rho}_{lm}$, then $\hat{R}$ can be defined as

$$\hat{R} = [\hat{\rho}_{lm} I(p_{lm} < \alpha)]_{l,m=1,\ldots,d}, \tag{13}$$

where α is a pre-specified significance level, e.g. $\alpha = 0.005$. We denote the $\hat{R}$ obtained based on p-values as "SECOM (Filtering)".

Estimation of nonlinear correlation coefficient

In addition to linear correlation, we are also interested in describing nonlinear correlations among taxa within an ecosystem (e.g., gut or oral environments) or across ecosystems. Therefore, the concept of distance correlation[36] was adopted to quantify general dependence between a pair of taxa.

Definition 0.1 (Distance correlation coefficient)**.** Let X and Y denote two random variables. Then the distance covariance and correlation coefficient between X and Y are defined as following

(1) $dCov^2(X,Y) = \frac{1}{\pi^2} \iint \frac{\|f_{X,Y}(t,s) - f_X(t) f_Y(s)\|^2}{t^2 s^2} dtds$,

(2) $dVar^2(X) = \frac{1}{\pi^2} \iint \frac{\|f_{X,X}(t,s) - f_X(t) f_X(s)\|^2}{t^2 s^2} dtds$,

(3) $dCor(X,Y) = \frac{dCov(X,Y)}{\sqrt{dVar(X) dVar(Y)}}$.

where $\|\cdot\|$ denotes Euclidean norm, and

(1) $0 \leq dCor(X,Y) \leq 1$,

(2) $dCor(X,Y) = 0$ if and only if X and Y are independent,

(3) $dCor(X,Y) = 1$ implies that dimensions of the linear subspaces spanned by X and Y samples respectively are almost surely equal.

Estimator of the distance correlation coefficient based on random samples $X = (X_1, X_2, \ldots, X_n)'$ and $Y = (Y_1, Y_2, \ldots Y_n)'$ is derived as follows. Define pairwise distances as $d_{ij} = |x_i - x_j|, e_{ij} = |y_i - y_j|$, and doubly centered distances as $D_{ij} = d_{ij} - \bar{d}_{i\cdot} - \bar{d}_{\cdot j} + \bar{d}_{\cdot\cdot}, E_{ij} = e_{ij} - \bar{e}_{i\cdot} - \bar{e}_{\cdot j} + \bar{e}_{\cdot\cdot}, i \neq j$, the empirical estimates of distance covariance and correlation can be obtained by

(1) $dCov_n^2(X,Y) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n D_{ij} E_{ij}$,

(2) $dVar_n^2(X,Y) = \frac{1}{n^2}\sum_{i=1}^n \sum_{j=1}^n D_{ij}^2$.

Note that the rank-transformed data also preserve the general dependence in the original data. We illustrate this using both linear and quadratic examples as shown in Supplementary Fig. 5a and b, respectively. In Supplementary Fig. 5a, $y = 2x + e, x \sim U[-2,2], e \sim N(0,1)$; whereas in Supplementary Fig. 5b, $y = x^2 + e, x \sim U[-2,2], e \sim N(0,1)$. It is easy to see from the figure that even though the shape of (X,Y) is not perfectly recovered using ranks (denoted as $(r(X), r(Y))$), distance correlations computed using the original data and the rank-transformed data (denoted as $dCor(X,Y)$ and $dCor(r(X), r(Y))$, respectively) are close enough to each other with a minimal numerical difference, which means the distance correlation computed from the rank-transformed data provides a reasonably good approximation for the distance correlation computed from the original data. Therefore, researchers can use either original abundances or the rank-transformed abundances to calculate distance correlations.

Similar to the procedure shown in the estimation of linear correlation coefficients, define

$$
\begin{aligned}
&d_{lm} := dCor(y_{il}, y_{im}) \quad (\text{or } dCor(r_{il}, r_{im})),\\
&\hat{d}_{lm} := dCor(\hat{y}_{il}, \hat{y}_{im}) \quad (\text{or } dCor(\hat{r}_{il}, \hat{r}_{im})),\\
&D = [d_{lm}]_{l,m=1,\ldots,d},\\
&D_n = [\hat{d}_{lm}]_{l,m=1,\ldots,d},
\end{aligned}
$$

the sparse distance correlation matrix estimation can be obtained by thresholding or p-values:

$$\hat{D} = sign(D_n)(|D_n| - \lambda)_+, \text{ or} \tag{14}$$

$$\hat{D} = [\hat{d}_{lm} I(p_{lm} < \alpha)]_{l,m=1,\ldots,d}. \tag{15}$$

Data availability

Simulation data can be found under https://github.com/FrederickHuangLin/SECOM-Code-Archive. The NoMIC data used in this study are not publicly available but may be obtained by contacting Dr. Merete Eggesbø (Merete.Eggesbo@fhi.no).

Code availability

All analyses can be found under https://github.com/FrederickHuangLin/SECOM-Code-Archive.

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Author contributions

Theory and Methodology were conceived and developed by HL and SDP. Illustration section was developed by HL, ME and SDP. Interpretation of the results from the analysis of NoMIC study were performed by ME and SDP. All numerical works were carried out by HL and ME. All three authors contributed towards writing the paper.

Competing interests

Authors declare no conflicts of interest.

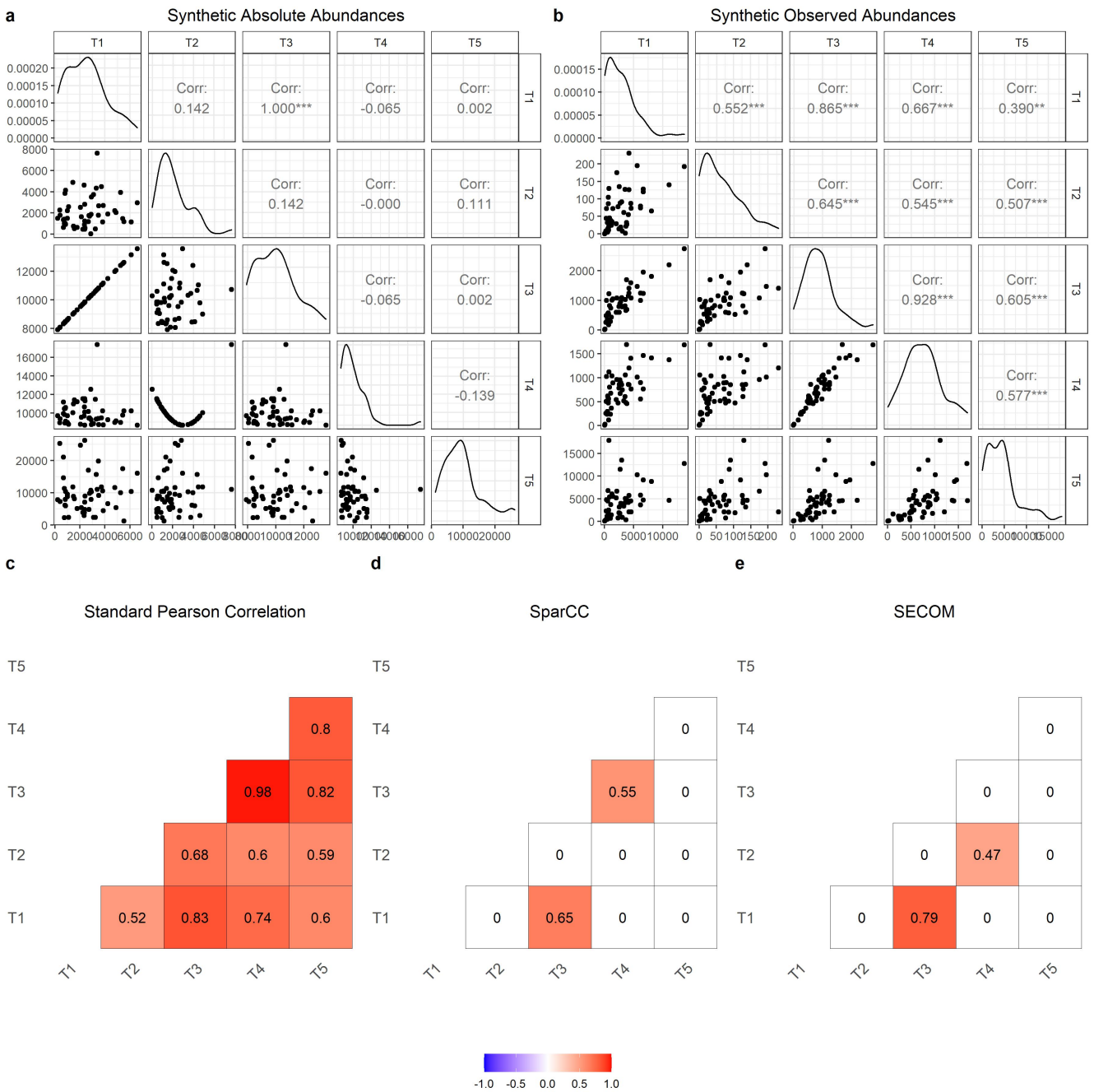

Figure 1. A toy example showing the difference between correlation methods. **a** A total of 100 taxa were generated, where the true abundances in the ecosystem were generated using a Negative-Binomial distribution. Taxa T1 and T3 are perfectly linearly correlated, and taxa T2 and T4 are quadratically correlated. The remaining taxa are uncorrelated. Only the first 5 taxa (T1 to T5) are provided for the simplicity of exposition. The lower triangular part of the matrix shows the pairwise scatter plots, the upper triangular part of the matrix shows the corresponding pairwise Pearson correlation coefficients, and the diagonal of the matrix shows the density plots for each taxon. **b** Observed abundances were generated by incorporating sample-specific sampling fractions and taxon-specific sequencing efficiencies to the true abundances. The lower triangular part of the matrix shows the pairwise scatter plots, the upper triangular part of the matrix shows the corresponding pairwise Pearson correlation coefficients, and the diagonal of the matrix shows the density plots for each taxon. **c**, **d**, **e** Pairwise correlation coefficients returned from the standard Pearson correlation coefficients, SparCC, and SECOM based on the observed abundances, respectively. The correlation coefficient ranges from -1 to 1, color coded by blue to red, correspondingly.

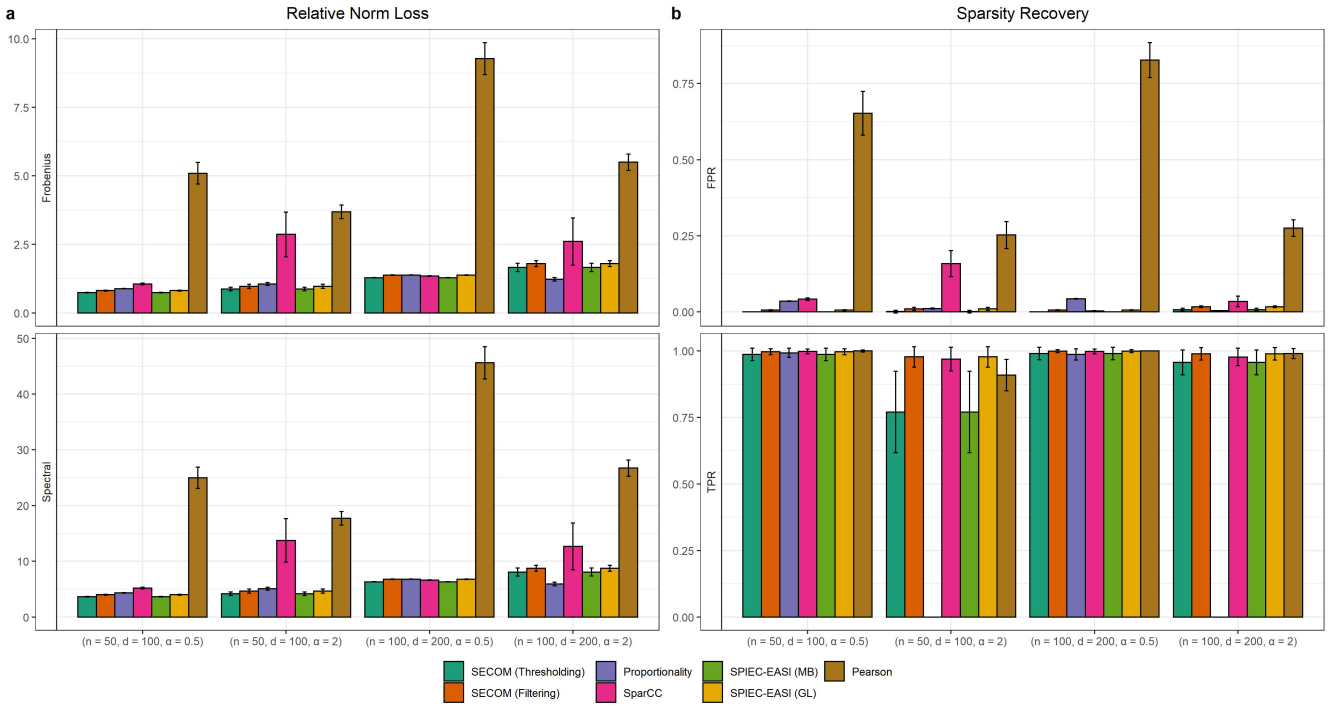

Figure 2. Comparisons of estimation accuracy and sparsity recovery for linear relationships within an ecosystem. The average relative norm loss (Frobenius and Spectral) and sparsity recovery (FPR/TPR) of various correlation methods are shown in **a** and **b**, respectively. Synthetic data were generated from Negative-Binomial (NB) distributions. The X-axis denotes the simulation settings, which are a combination of sample size n, number of taxa d, and the dispersion parameter α of the corresponding NB distribution. The number of iterations for each simulation setting = 100. Data are represented by the mean of all iterations and error bars represent standard errors. Color and the name of the corresponding correlation methods are shown at the bottom within the graph. The results demonstrate that SECOM (Thresholding and Filtering) and SPIEC-EASI (MB and GL) outperformed all existing methods not only in terms of estimation accuracy, but also in terms of uniformly small FPR, and comparable TPR.

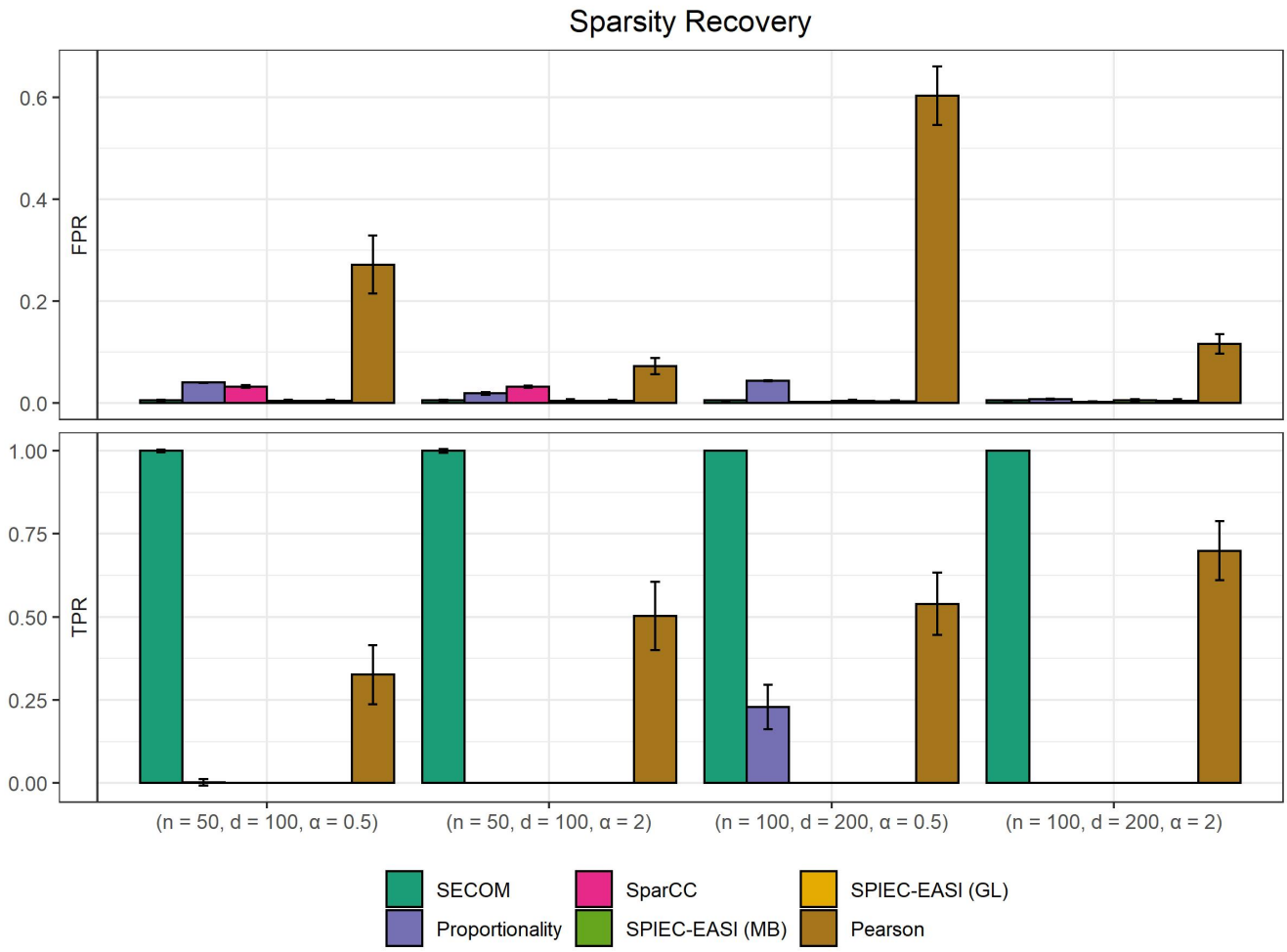

Figure 3. Comparisons of sparsity recovery for nonlinear relationships within an ecosystem. Synthetic data were generated from Negative-Binomial (NB) distributions. The X-axis denotes the simulation settings, which are a combination of sample size n, number of taxa d, and the dispersion parameter α of the corresponding NB distribution. The number of iteration for each simulation setting = 100. Data are represented by the mean of all iterations and error bars represent standard errors. Color and the name of the corresponding correlation methods are shown at the bottom within the graph. Results showed that only SECOM controlled the FPR while maintaining high TPR. Other than SECOM, none of existing methods identify nonlinear relationships.

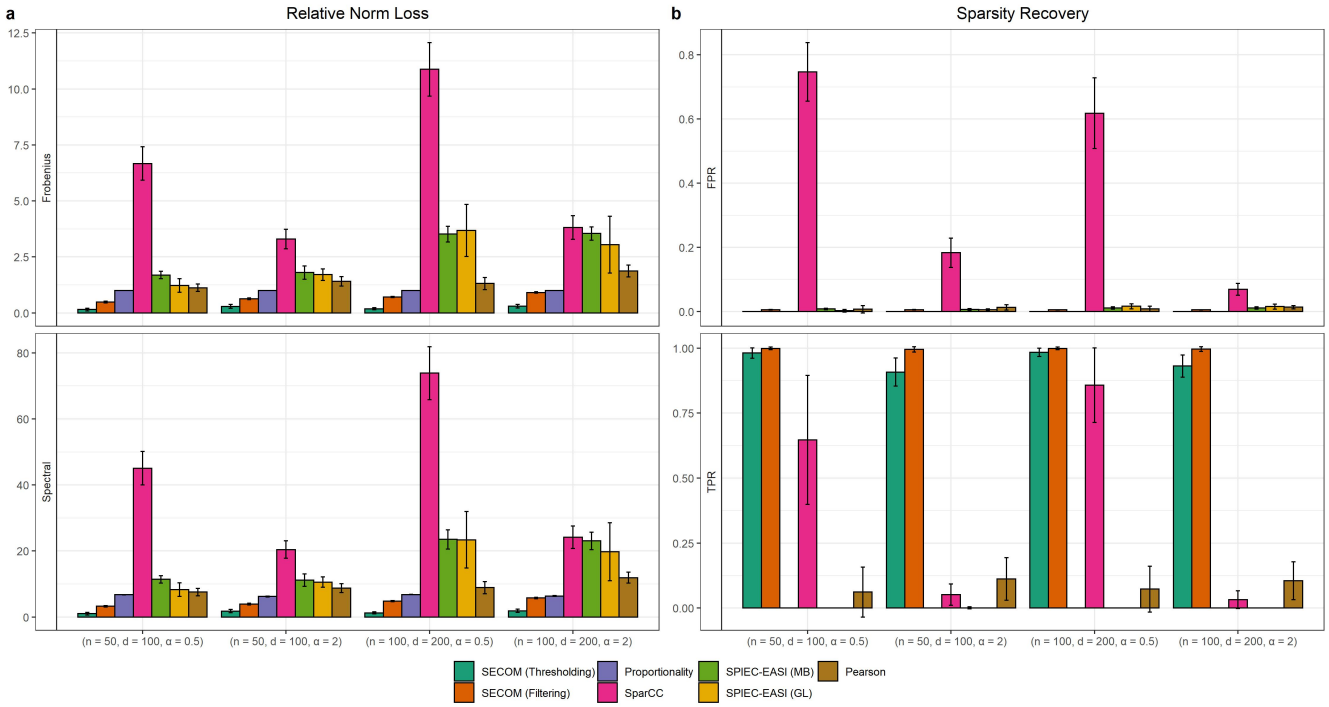

Figure 4. Comparisons of estimation accuracy and sparsity recovery for linear relationships between two ecosystems. The relative norm loss (Frobenius and Spectral) and sparsity recovery (FPR/TPR) of various correlation methods are shown in **a** and **b**, respectively. Synthetic data were generated from Negative-Binomial (NB) distributions. The X-axis denotes the simulation settings, which are the combinations of sample size n, number of taxa d, and the dispersion parameter α of the corresponding NB distribution. Data from different ecosystems have different sampling fractions and sequencing efficiencies in each simulation run. The number of iterations for each simulation setting = 100. Data are represented by the mean of all iterations and error bars represent standard errors. Color and the name of the corresponding correlation methods are shown at the bottom within the graph. Results demonstrated that SECOM methods, namely, SECOM(Thresholding) and SECOM(Filtering) were the only methods successfully quantified microbial correlations between ecosystems.

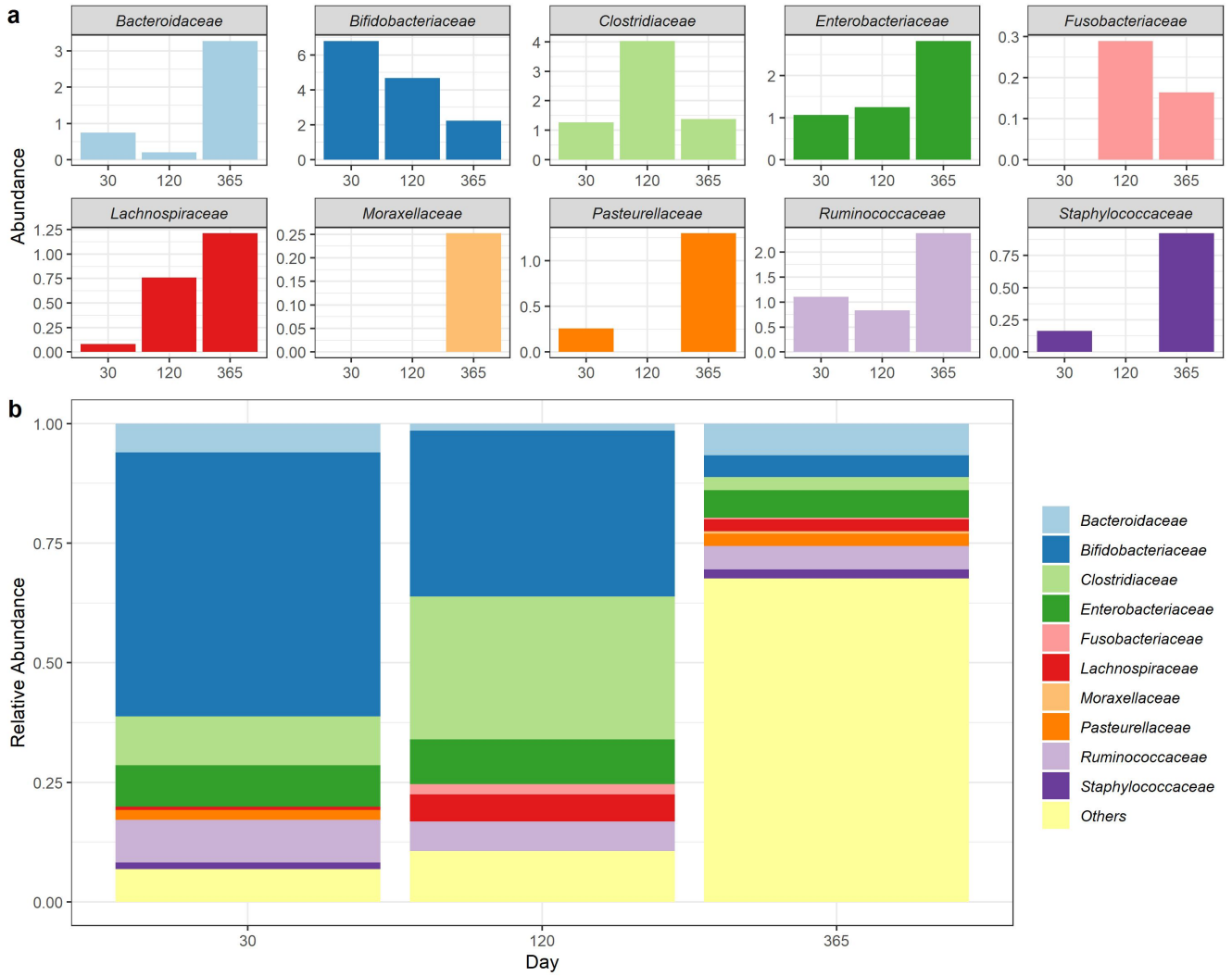

Figure 5. Temporal patterns of bias corrected abundance (**a**) and relative abundances (**b**) for Norwegian infant gut microbiome data among families. The x-axis represents days (30, 120, and 365 days) and y-axis denotes either the bias corrected abundances or relative abundances. The top 10 most abundant taxa up to 365 days were selected for the visualization purpose. Families with prevalence across samples $< 10\%$ were not shown separately and grouped into "Others". Different taxa were coded by different colors as shown on the legends.

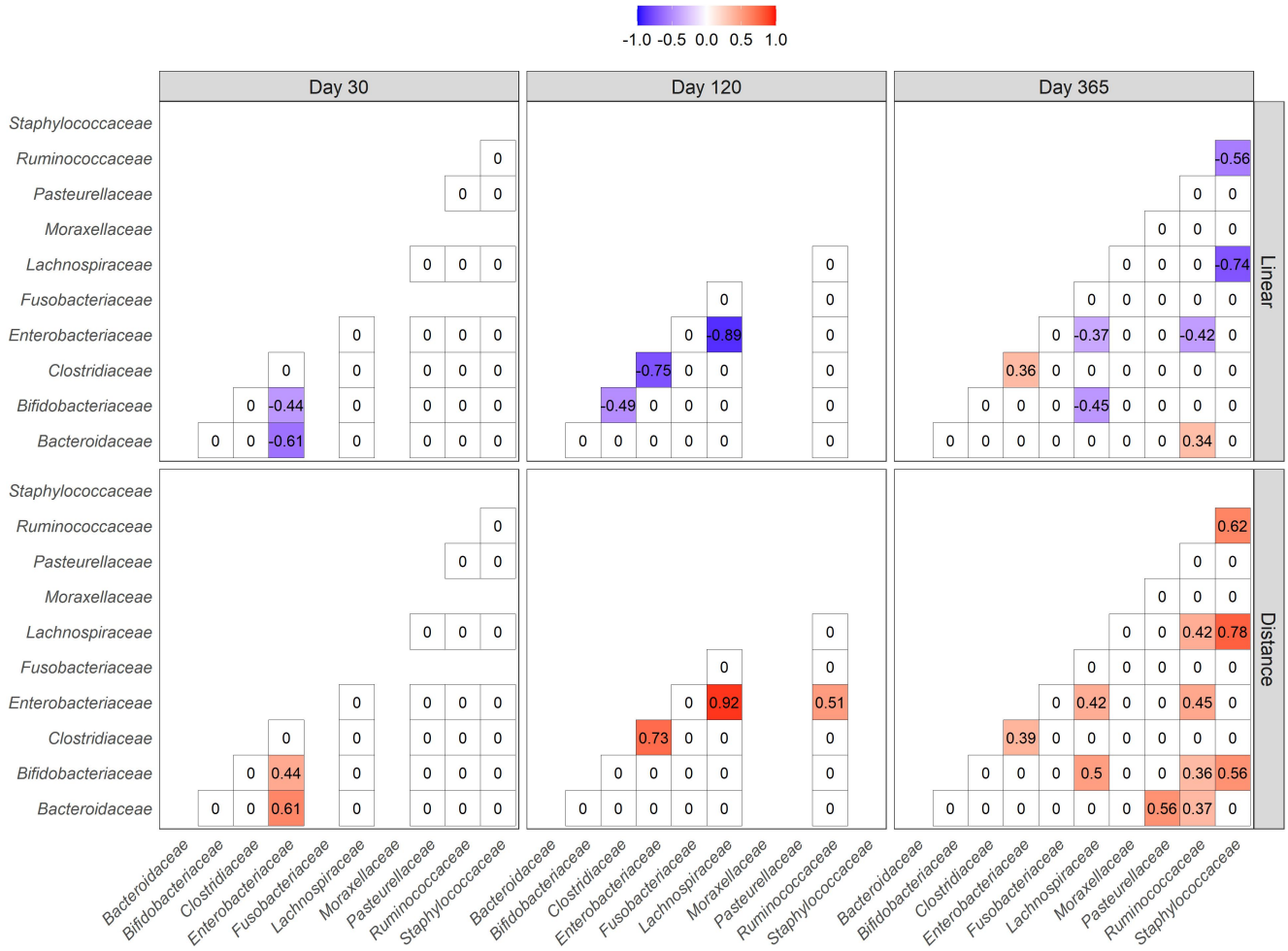

Figure 6. Temporal patterns of correlations for Norwegian infant gut microbiome data among families. The correlation metrics (Linear and Distance) are represented by rows and days (30, 120, and 365 days) are represented by columns, respectively. The top 10 most abundant families up to 365 days were initially selected for the visualization purpose. Families with prevalence across samples $< 10\%$ were discarded and left blank in the heatmap. The linear correlation coefficient ranges from -1 to 1, color coded by blue to red, respectively.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SECOMSupplementaryInformation.pdf
- suppfig1.pdf
- suppfig2.pdf
- suppfig3.pdf
- suppfig4.pdf
- suppfig5.pdf